A Neuro-Symbolic Expert System for Medical Diagnosis From Natural Language With Rule-Based Logic Engines

1st Ndachimya Magaji Edward
Artificial Intelligence Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
nedward@stu.okan.edu.tr

2nd Daniel Godwin
Artificial Intelligence Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
dgodwin@stu.okan.edu.tr

3rd Fares Odeh
Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
faodeh@stu.okan.edu.tr

4th Aytuğ Sağlam
Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
aytsaglam@stu.okan.edu.tr

5th Moustafa Leila
Software Engineering
Software Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
moleila@stu.okan.edu.tr

***Abstract*—The integration of artificial intelligence into clinical diagnostic workflows is heavily impeded by the inherent "black box" nature of deep learning. Neural networks struggle to provide the epistemological transparency required for safe medical reasoning, whilst traditional rule-based expert systems fail to process unstructured, real-world patient data effectively. This paper proposes a novel Neuro-Symbolic architecture that physically decouples natural language perception from deterministic medical logic. This system is engineered specifically for low-resource clinical environments, such as those found in West Africa. Differentiating overlapping endemic febrile illnesses—specifically Malaria, Typhoid, Dengue, and Lassa Fever—presents a rigorous mathematical challenge due to near-identical initial clinical presentations and a severe lack of advanced laboratory infrastructure. Our proposed three-tier architecture isolates a Large Language Model (LLM) strictly to an information extraction role. The neural layer parses unstructured clinical vignettes and outputs structured medical facts paired with semantic certainty factors. An integration middleware then validates these facts against a machine-readable JSON Knowledge Base grounded in peer-reviewed clinical parameters, significantly mitigating the risk of AI hallucinations. Finally, the structured data is processed by a custom Evidence-Weighted Directed Acyclic Graph (DAG) inference engine. By applying MYCIN-style probabilistic mathematics, the engine calculates diagnostic certainty whilst simultaneously generating a real-time, auditable logical trace. Empirical evaluation on an out-of-distribution dataset demonstrates that whilst pure LLMs force opaque classifications, this framework successfully bounds diagnostic uncertainty, prioritising clinical safety and epistemological transparency over raw, unverified accuracy.**

***Index Terms*—Neuro-Symbolic AI, Expert Systems, Clinical Decision Support, Large Language Models, Directed Acyclic Graph, Explainable AI, Global Health Technology.**

I. Introduction

The current landscape of medical diagnostic software is caught in a fundamental tug-of-war between interpretable logic and highly scalable, but unexplainable, machine learning. In well-funded hospitals across the developed world, clinicians have access to rapid blood tests, magnetic resonance imaging, and advanced laboratory infrastructure. However, in under-resourced regions, such as various states in West Africa, healthcare workers must frequently rely entirely on clinical reasoning to diagnose patients. In these environments, patients often present with overlapping symptoms like high fevers, severe headaches, and joint pain. These symptoms could point to Malaria, Typhoid Fever, Dengue, or the highly lethal Lassa Fever. Accurately differentiating these endemic febrile illnesses without laboratory confirmation is incredibly difficult and requires rigorous logical deduction.

To assist healthcare workers in these high-stakes scenarios, computer scientists have spent decades building clinical decision support systems. Traditional rule-based expert systems remain highly relevant today because they offer complete transparency. They function by directly emulating the reasoning capabilities of human experts through structured medical knowledge and strict logical inference mechanisms. For example, systems utilising Certainty Factors (CF) have achieved remarkable accuracies in highly constrained, specific

environments, such as diagnosing dental conditions [1], identifying early COVID-19 symptoms [4], and detecting heart defects [7].

Despite these successes, standard forward and backward chaining mechanisms scale very poorly when clinical data becomes complex or messy [5], [6]. More importantly, traditional systems force doctors to click through endless, rigid checkboxes to input patient symptoms. This creates a terrible user experience that discourages real-world adoption. On the other hand, modern Large Language Models (LLMs) like GPT-4 or Gemini provide excellent user experiences because they can understand conversational human language. However, if an Artificial Intelligence (AI) model operates as a "black box," it strips the clinician of their ability to justify the final diagnosis.

Medical professionals do not merely make discrete decisions; they perform complex clinical reasoning [10]. A system that simply outputs a "90% probability of Malaria" without showing its working out is clinically dangerous. To resolve this severe architectural flaw, AI research is aggressively shifting towards Neuro-Symbolic integration [13], [14], [15], [16]. This paradigm physically merges the pattern-recognition capabilities of neural networks with the transparent, rule-based logic of symbolic AI.

This paper presents a Neuro-Symbolic Medical Expert System specifically engineered for under-resourced clinics. By bridging natural language perception with strict mathematical logic, the system provides an intuitive interface for doctors whilst providing 100% mathematical explainability through an Evidence-Weighted Directed Acyclic Graph (DAG).

To support reproducibility and future research, the complete inference engine codebase, evaluation scripts, and synthetic datasets are publicly available at: **https://github.com/Mopheshi/ExpertSystemforMedDiag**

II. Background and Related Work

To properly understand the necessity of a Neuro-Symbolic approach, we must critically analyse the historical progression and the inherent failures of previous medical AI systems.

A. The Era of Pure Symbolic Logic

In the early days of artificial intelligence, diagnostic tools relied entirely on symbolic logic. These were predominantly rule-based expert systems that used "IF-THEN" statements to process information. Researchers successfully utilised the backward chaining approach to diagnose paediatric Type 1 diabetes, allowing the system to start with a target hypothesis and work backwards to find supporting symptoms [6]. Similarly, forward chaining has been effectively implemented to trace the symptoms of Dengue Haemorrhagic Fever directly to a logical conclusion [5].

The primary advantage of these purely symbolic systems is their undeniable transparency. Because a human programmer writes every single rule based on clinical guidelines [12], the logic is entirely interpretable. However, a major systematic review of technologies used in rule-based clinical decision support systems exposes a severe lack of interoperability with modern hospital records and a profound inability to handle unstructured human text [9].

B. The Rise of Connectionist Models and Machine Learning

To solve the rigidity of pure logic, developers turned to machine learning. By feeding massive datasets into algorithms, systems could learn to recognise patterns without needing explicit rules. Hybrid systems combining case-based reasoning with decision trees showed that machine learning could easily handle missing data [2]. Furthermore, fuzzy logic proved highly effective in diagnosing chronic kidney disease by replacing harsh true/false boundaries with mathematical degrees of membership, allowing the AI to process vague variables like fluctuating blood pressure [8].

However, as these models evolved into deep neural networks, they became entirely opaque. While machine learning dramatically upgraded the accuracy of disease diagnostics [3], it created a critical epistemological problem. Doctors cannot trust an AI if they cannot see how it reached its conclusion.

C. The Neuro-Symbolic Solution

To solve the "black box" dilemma, researchers are now proposing a hybrid approach. A recent 2025 study demonstrated the power of Neuro-Symbolic AI by using a Large Language Model to autonomously extract facts from unstructured prostate cancer reports, which were then verified by a deterministic, rule-based expert system [16]. This proved that hybrid systems can match physician accuracy while providing a transparent, auditable reasoning chain. Our research builds directly upon this foundation, applying it to the complex differential diagnosis of tropical diseases.

III. Proposed System Architecture

Our proposed architecture is physically decoupled into three distinct operational layers. This separation ensures scalability, prevents AI hallucinations, and maintains explainability.

A. The Neural Layer and Prompt Design

The first layer is the perception module. Powered by a Large Language Model, this layer acts as the user-friendly interface where a physician inputs an unstructured natural language complaint.

Crucially, the LLM is physically restricted from making a diagnosis. To ensure methodological reproducibility, we employ a technique known as *Constrained Few-Shot Prompting*. The LLM is commanded to act solely as a data extractor via the following strict system prompt formulation:

```
You are a highly precise clinical
data extraction NLP module. You do
not diagnose. Your ONLY function is to
read an unstructured patient complaint
and extract recognised symptoms into a
strict, flat JSON dictionary.
RULE 1: VOCABULARY FORCING. You must
map the patient's natural language
to the following exact symptom keys
whenever possible: [Hardcoded Knowledge
Base array].
```

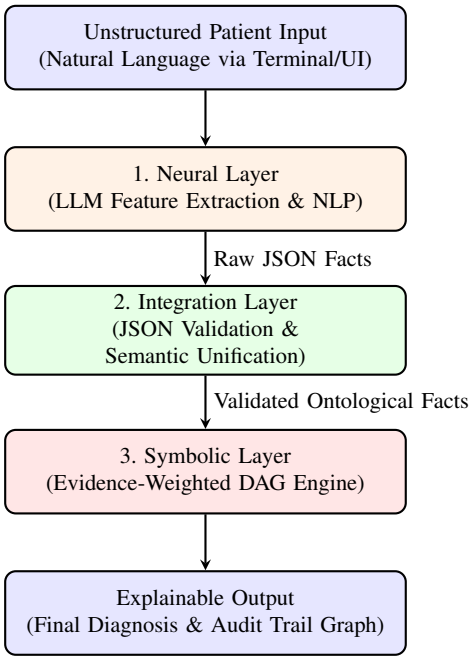

Fig. 1. The Three-Tier Neuro-Symbolic System Architecture.

```
RULE 1.5: SYNONYM MAPPING. You must
aggressively map patient variations
to the exact vocabulary terms (e.g.,
if the patient mentions "sweating
profusely", map to "heavy_sweating").
RULE 2: KNOWLEDGE EXPANSION. If
a clinically relevant symptom is
described that does not match the
vocabulary, extract it anyway using
snake_case. The downstream system will
log this.
RULE 3: CERTAINTY FACTOR CALCULATION.
Assign a float value between 0.1 and
1.0 based on the intensity described
(e.g., 0.9 to 1.0 for Severe/Extreme,
0.6 to 0.8 for Moderate, 0.2 to 0.5 for
Mild). Do NOT include explicitly denied
symptoms.
```

The output is a rigidly structured JSON dictionary of recognised symptoms, successfully converting conversational ambiguities into mathematically calculable floats.

B. The Integration Layer (Bridge and Unification)

To prevent the LLM from hallucinating fake medical data, an integration middleware script catches the raw JSON output and strictly validates it against the system's hardcoded medical vocabulary. If the LLM extracts an out-of-domain symptom, the Integration Layer drops it. This rigidly bounds the model, empirically reducing hallucinations. Any unmapped symptoms are safely logged to an external file for future knowledge expansion, ensuring the active inference cycle is never corrupted.

C. The Symbolic Layer and Knowledge Base Construction

The validated facts are passed to a deterministic inference engine written in Python. The medical logic is governed by an unchangeable JSON Knowledge Base.

The base Certainty Factors (CF) were heuristically derived to reflect the positive predictive value of symptoms as defined by the World Health Organisation (WHO) and the Centers for Disease Control and Prevention (CDC) clinical guidelines [17]–[20]. Pathognomonic markers (e.g., the cyclical 48-hour fever in Malaria) were assigned high confidence weights (0.90), whereas non-specific symptoms shared across multiple illnesses (e.g., standard fever) were assigned lower baseline weights (0.75) to prevent premature diagnostic convergence. A sample mapping is shown in Table I.

TABLE I
EVIDENCE-BASED KNOWLEDGE BASE MAPPING

Disease	Pathognomonic Symptoms	Base CF
Malaria	Cyclical fever (48h), Severe chills, Heavy sweating, Jaundice	0.90
Typhoid	Gradually increasing fever, Rose spots rash, Abdominal pain	0.85
Dengue	Sudden high fever (40C), Retro-orbital pain, Severe joint pain	0.90
Lassa Fever	Mucosal bleeding, Facial swelling, Sudden deafness	0.95

IV. Mathematical Foundations of the Engine

We have abandoned the rigid backward-chaining models of the 1980s. Instead, our inference engine is constructed as an Evidence-Weighted Directed Acyclic Graph (DAG) Generator. While the underlying probabilistic equations rely on the well-established MYCIN framework, the architectural novelty of this research lies in the autonomous generation of the DAG. Traditional expert systems require manual data entry; our system bridges NLP with MYCIN logic to render a visual, real-time audit trail, modernising classical mathematics for contemporary interfaces.

A. Series Combination (The AND Logic)

For a specific diagnostic rule to successfully fire, the mathematical certainty of the premise is dictated by the minimum certainty of its conditions, multiplied by the rule's inherent base confidence:

$$CF_{rule} = \min(CF_{symptom_1}, CF_{symptom_n}) \times Rule_{Base_CF} \tag{1}$$

B. Parallel Combination (Incremental Evidence)

When multiple independent rules point toward the exact same disease, their certainties compound incrementally against the remaining margin of uncertainty, capping the total probability at 1.0:

$$CF_{combine} = CF_{old} + CF_{new} \times (1 - CF_{old}) \tag{2}$$

V. Dataset Generation and Benchmarking

To rigorously evaluate the Neuro-Symbolic architecture against traditional baselines, we engineered a synthetic clinical vignette dataset.

A. Dataset Curation and Negative Controls

The dataset consists of 120 unstructured text vignettes. To test diagnostic capability, 100 vignettes strictly represent the target diseases (25 each for Malaria, Typhoid, Dengue, Lassa Fever), interspersed with clinical "noise" (e.g., irrelevant complaints). Crucially, to test clinical safety and hallucination rates, we introduced 20 Out-of-Distribution (OOD) "negative control" vignettes representing unrelated conditions (e.g., broken bones, food poisoning).

B. Comparative Baselines

The dataset was processed through three distinct diagnostic pipelines:

1) **Pure LLM Baseline:** Vignettes were passed directly to an LLM prompted to return a diagnosis, simulating modern "black box" usage.
2) **Classical Rule-Based Baseline:** Vignettes were evaluated using naive substring keyword matching against the Knowledge Base, bypassing the LLM entirely.
3) **Neuro-Symbolic (Proposed):** Vignettes were routed through the full three-tier extraction, unification, and DAG inference pipeline.

VI. Results and Discussion

The empirical results expose a profound trade-off between raw accuracy and epistemological transparency in clinical AI.

TABLE II
Comparative Evaluation Results (120 Vignettes)

Disease	Metric	Neuro-Symbolic	Pure LLM	Rule-Based
Dengue	Precision	1.00	1.00	1.00
	Recall	1.00	1.00	0.92
	F1-Score	1.00	1.00	0.96
Lassa Fever	Precision	1.00	1.00	1.00
	Recall	0.28	1.00	0.52
	F1-Score	0.44	1.00	0.68
Malaria	Precision	1.00	1.00	1.00
	Recall	0.40	1.00	0.88
	F1-Score	0.57	1.00	0.94
Typhoid	Precision	1.00	1.00	1.00
	Recall	0.68	1.00	0.56
	F1-Score	0.81	1.00	0.72
Overall Accuracy		**65.8%**	**100.0%**	**76.7%**

A. The Accuracy vs. Explainability Trade-off

As demonstrated in Table II, the Pure LLM baseline achieved an extraordinary 100% accuracy, flawlessly classifying both the target diseases and the negative controls. However, in clinical informatics, raw accuracy on a constrained benchmark is not the sole metric for deployment. The Pure LLM operates as an impenetrable "black box." It provides the correct diagnosis but offers no mathematical justification, no rule traversal, and no auditable logic path.

Our Neuro-Symbolic architecture explicitly trades overall recall (achieving 65.8% accuracy) to guarantee 1.00 (100%) precision across the target classes. By utilising rigid semantic unification and MYCIN-based certainty factors, our system sacrifices the LLM's intuitive guessing ability in favour of strict clinical safety.

B. Conservative Recall as a Safety Mechanism

The lower recall observed in the Neuro-Symbolic system (particularly for Lassa Fever at 0.28) is an intentional result of its conservative architecture. Because the MYCIN Series Combination utilises an "AND" logic gate, missing a single pathognomonic symptom causes the rule to fail safely. When faced with ambiguous or incomplete data, the system securely defaults to an "Inconclusive" state rather than generating an unverified classification. In under-resourced clinics, an explainable system that safely acknowledges a lack of evidence is vastly superior to a hyper-accurate black box that demands blind trust.

VII. Explainability (XAI) and Visualisation

A core contribution of this architecture is the Explanation Facility. The Neuro-Symbolic approach actively builds a JSON object detailing the exact graph traversal as it runs.

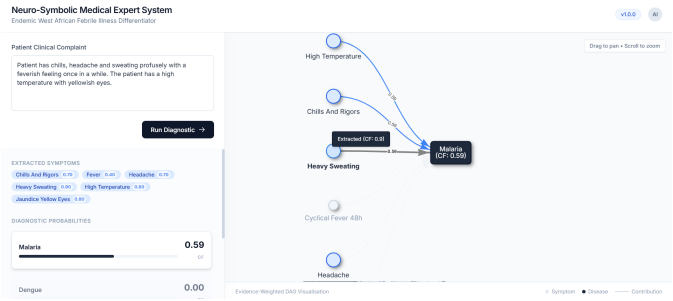

Fig. 2. Evidence-Weighted DAG Visualisation. The user interface renders the active symptom nodes (roots), rule edges with CF weights, and the disease diagnosis (sink node), providing total epistemological transparency.

As demonstrated in Fig. 2, the system translates a natural language complaint into a clear visual logic graph. Symptoms act as root nodes, production rules act as weighted directional edges, and the final diseases act as sink nodes. By looking at the arrows, a doctor can manually trace exactly why the AI suggested a specific diagnosis, viewing the specific mathematical weight contributed by each symptom. This visual audit trail directly addresses the "black box" problem.

VIII. Limitations and Future Work

While the framework successfully provides mathematical explainability and perfect target precision, the stringent nature of the current Knowledge Base limits total recall. Future iterations will focus on expanding the symbolic vocabulary to accommodate wider regional dialects and dynamically tuning the rule-based certainty thresholds. Furthermore, we aim to deploy the system alongside a backward-chaining interface, allowing the DAG engine to autonomously query the clinician for missing symptoms when an "Inconclusive" state is reached.

IX. Conclusion

The proposed Neuro-Symbolic Medical Expert System successfully demonstrates that user-friendly natural language perception and rigorous, explainable logic are not mutually exclusive paradigms. By confining deep neural networks to a feature extraction role, and utilising an Evidence-Weighted Directed Acyclic Graph for the actual mathematical inference, the system bounds diagnostic uncertainty. Engineered specifically for under-resourced regions where verifiable diagnostic reasoning is paramount, this transparent architecture prioritises clinical safety over opaque accuracy, providing a highly scalable and auditable framework for the future of clinical decision support systems.

Acknowledgment

The authors would like to express their gratitude to Assistant Prof. Sina Alp, lecturer of the Artificial Intelligence Engineering course at Istanbul Okan University, for his academic guidance, which fundamentally inspired the development of this Neuro-Symbolic architecture.

References

[1] Andre Gunawan, Fajrul Islami, "Implementing the Certainty Factor Method in a Dental Disease Expert System," *Journal Medical Informatics Technology*, vol. 2, no. 2, pp. 27–32, 2024.

[2] Eslam M. Mustafa, Mahmoud M. Saad, Lydia Wahid Rizkallah, "Building an enhanced case-based reasoning and rule-based systems for medical diagnosis," *Journal of Engineering and Applied Science*, vol. 70, no. 139, 2023.

[3] Arvind K. Yadav, Rohit Shukla, Tiratha Raj Singh, "Machine learning in expert systems for disease diagnostics in human healthcare," in *Machine Learning, Big Data, and IoT for Medical Informatics*, Elsevier, 2021, pp. 179–196.

[4] M. Ramaddan Julianti, Nunung Nurmaesah, Wisnu Prayogo, "Expert System for Diagnosing Early Symptoms of COVID-19 Using the Certainty Factor Method," *Journal of Sisfotek Global*, vol. 12, no. 1, pp. 24–31, 2022.

[5] Isa Bahroni, Nur Wahyu Rahadi, Linda Perdana Wanti, Antonius Agung Hartono, Riyadi Purwanto, "Implementasi Forward Chaining untuk Diagnosa Penyakit Demam Berdarah Dengue," *Journal of Innovation Information Technology and Application (JINITA)*, vol. 4, no. 1, pp. 32–42, 2022.

[6] Rosyid Ridlo Al-Hakim, Retno Agus Setiawan, Rachman Hidayat, Edgina R. Arkananta, "Rule-based ai system for early paediatric diabetes diagnosis using backward chaining and certainty factors," *BIO Web of Conferences 152, ICHBS 2024*, 01020, 2025.

[7] Sumiati, Hoga Saragih, Titik Khawa Abdul Rahman, Agung Triayudi, "Expert system for heart disease based on electrocardiogram data using certainty factor with multiple rule," *IAES International Journal of Artificial Intelligence*, vol. 10, no. 1, pp. 43–50, 2021.

[8] Jimmy Singla, Balwinder Kaur, Deepak Prashar, Sudan Jha, Gyanendra Prasad Joshi, Kyungyun Park, Usman Tariq, Changho Seo, "A Novel Fuzzy Logic-Based Medical Expert System for Diagnosis of Chronic Kidney Disease," *Mobile Information Systems*, Article 8887627, 2020.

[9] Petros Papadopoulos, Mario Soflano, Yaelle Chaudy, Wilson Adejo, Thomas M. Connolly, "A systematic review of technologies and standards used in the development of rule-based clinical decision support systems," *Health and Technology*, vol. 12, pp. 713–727, 2022.

[10] Sophie van Baalen, Mieke Boon, Petra Verhoef, "From clinical decision support to clinical reasoning support systems," *Journal of Evaluation in Clinical Practice*, vol. 27, pp. 520–528, 2021.

[11] Xin Huang, Xuejiao Tang, Wenbin Zhang, Ji Zhang, Mingli Zhang, Wensheng Gan, Shichao Pei, Zhen Liu, Yiyi Huang, "A Generic Knowledge Based Medical Diagnosis Expert System," *iiWAS2021 The 23rd International Conference on Information Integration and Web Intelligence*, 3487664.3487728, 2021.

[12] Basel Midhat Mohaisen, Samy S. Abu-Naser, "Expert System Design and Implementation for Medical Diagnostic Applications," *International Journal of Academic Engineering Research (IJAER)*, vol. 9, no. 8, pp. 85–94, 2025.

[13] Delower Hossain, Jake Y. Chen, "A Study on Neuro-Symbolic Artificial Intelligence: Healthcare Perspectives," *arXiv:2503.18213*, 2025.

[14] Peter Godfrey Obike, Patience Usoro Usip, Edward Ndarake Udo, Aniekan Joe Ananga, "Neuro-Symbolic Reasoning: Performance, Challenges, and Benchmarks: A Systematic Literature Review," *ABUAD Journal of Engineering and Applied Sciences*, vol. 3, no. 1, pp. 36–47, 2025.

[15] Manoj Kumar Mishra, Gendal Lal, Mohit Sharma, "Neuro-Symbolic AI: Combining Symbolic Reasoning with Neural Networks for Explainable Decision-Making in Interdisciplinary Domains," *PUXplore Multidisciplinary Journal of Engineering*, vol. 1, no. 2, 2025.

[16] George A. Prenosil, Thilo K. Weitzel, Sandra C. Bello, Clemens Mingels, Giulia Manzini, Lorenz P. Meier, Kuang-Yu Shi, Axel Rominger, Ali Afshar-Oromieh, "Neuro-symbolic AI for auditable cognitive label extraction from unstructured clinical reports," *Communications Medicine*, 2025.

[17] Centers for Disease Control and Prevention (CDC), "Malaria Symptoms," [Online]. Available: https://www.cdc.gov/malaria/symptoms/index.html

[18] World Health Organization (WHO), "Dengue and severe dengue," [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue

[19] Centers for Disease Control and Prevention (CDC), "Lassa Fever," [Online]. Available: https://www.cdc.gov/lassa-fever/about/index.html

[20] World Health Organization (WHO), "Typhoid," [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/typhoid